Sidharth Anupkrishnan

(413) 309-0534 | sanupkrishna@umass.edu | LinkedIn

Summary

PhD researcher at UMass Amherst developing multimodal wearable sensing systems using EEG, eye-tracking, and EDA to model human engagement and behavior. My work combines biosignal processing, human-subject studies, and machine learning for real-world applications.

Research Experience

Benchmarking EEG Foundation Models for Wearable EEG *(Ongoing work)* *August 2026 – Present*

- Benchmarking pretrained EEG foundation models on wearable EEG datasets to assess transfer feasibility and identify limitations of existing models.
- Evaluating robustness to reference shifts between mastoid or average referencing used in pretraining and the reference configurations of wearable EEG devices.
- Assessing sensitivity to electrode-position shifts between conventional scalp EEG and wearable sensor layouts to characterize model failure modes.

Fetal Health and Continuous Blood Pressure Monitoring *(Ongoing work)* *June 2026 – Present*

In Collaboration with Columbia, Dartmouth and Lumious Ultrasound

- Designing and fabricating an ear-based device for fetal heart health monitoring.
- Building a biosignal hub to acquire and preprocess data from multiple fetal monitoring devices, including ultrasound, an ECG belt, and the ear device.
- Developing continuous blood pressure monitoring using pulse arrival time (PAT) derived from in-ear photoplethysmography (PPG) and chest ECG.

Inter-Brain Synchrony using Wearable EEG (*Ongoing work)*

In Collaboration with Dolby Laboratories *March 2026 – Present*

- Designing an inter-brain synchrony monitoring platform using wearable EEG headbands to assess group-level synchrony as a proxy for classroom or group engagement**.**
- Conducting feasibility analysis of frontal and over-the-ear EEG channels, focusing on channel reliability, artifact sensitivity, and suitability for in-the-wild deployment.

E3Sense: Head-Confined Multimodal Sensing of Learner Engagement

(Co-first author, Submitted to ACM CHI)

In Collaboration with Dolby Laboratories *April 2025 – September 2026*

- Designed a head-mounted multimodal wearable integrating EEG, eye tracking, and forehead EDA to measure learner engagement.
- Conducted a 30-participant study collecting 450 engagement ratings and comprehension measures during educational videos.
- Developed participant-disjoint multimodal ML pipelines for ordinal engagement prediction and exploratory personalization using learners' definitions of engagement.
- Achieved 75.0% balanced within-one-level prediction versus a 63.0% majority-rating baseline on 15 held-out forehead-sensing participants.

Multimodal Engagement Decoding During Live Classroom Lectures

(Co-first author, Submitted to IEEE ICASSP; Poster presented at CNS 2026)

In collaboration with Dr. Lisa Sanders, Neurocognition and Perception Lab. *Sept 2025 – September 2026*

- Designed a 12-participant live-lecture study synchronizing a 64-channel scalp EEG system, wearable EEG headband, eye tracking, and instructor speech.
- Co-developed a multimodal CRNN combining headband EEG, gaze, and the speech envelope; achieved MAE 0.628 and 91.2% balanced within-one accuracy on held-out participants.
- Compared wearable and scalp EEG configurations and conducted speech-locked ERP analysis of N400 responses to LLM-identified unexpected words.

EMG-FoundBench: Benchmarking EMG Foundation Models
(Co-Author, Under review at ICLR 2027)
In collaboration with MoSIS Lab, University of Georgia *Jan 2026 – May 2026*

- Contributed to EMG-FoundBench, a unified benchmark spanning 20 public datasets, over 1 million EMG segments, and nine pretrained foundation models.
- Implemented standardized EMG preprocessing and evaluation for linear probing, full fine-tuning, cross-user generalization, few-shot adaptation, and sequence-to-sequence decoding.
- Evaluated transfer and personalization across heterogeneous EMG hardware, classification tasks, and continuous EMG-to-text decoding.

Omnibuds Earable Device
Nokia Bell Labs, Cambridge, UK *June 2024 – Aug 2024*

- Designed and developed adaptive load balancing strategy for stateful applications in ear based wearable devices for optimal vital signs monitoring using PPG, IMU and audio playback user experience.
- Experimented with model partitioning (TinyML) across earbuds and collaborative inference.
- Tested our implementation on OmniBuds earable platform. Patent was filed.

Design Optimization of Activity Recognition System on an Embedded Platform
Solarillion Foundation, Chennai, India *May 2017 – Apr 2018*

- Designed an HAR platform optimizing computational complexity, energy consumption and cost.
- Performed preprocessing, feature extraction on accelerometer, magnetometer, gyroscope data for activities.
- Optimized model performance on embedded devices by choosing minimal features and small ML models.

Education

University of Massachusetts, Amherst
PhD in Computer Science (Advised by Prof. VP Nguyen) *Sep 2023 – June 2028 (Expected Graduation)*
Master's in Computer Science *Sep 2021 – Aug 2023*
SRM Institute of Science and Technology
Bachelor's in Mechanical Engineering *Jul 2014 – July 2018*

Technical Skills

- **Programming & Analytics**: Python (NumPy, Pandas, Pytorch, MNE, Scikit, Tensorflow, Matplotlib, Seaborn), C, C++ (Unity), MatLab (EEGLAB), GoLang
- **Sensors/Devices**: Pupil Labs Neon Eye Tracker, Emotibit, Frenz EEG Headband, ESP32, OpenBCI Cyton, Omnibuds, XReal AR Glasses
- **Distributed Systems**: Docker, Kubernetes, Prometheus, Gluster, Ceph
- **Microcontrollers**: ESP32, MAX78000, Arduino

Industry Experience

Research Scientist Intern, Nokia Bell Labs (Ombibuds Earable Device) *June 2024 – Aug 2024*
Software Engineer, IBM (Distributed Storage Design for ML/HPC workloads) *June 2019 – Aug 2021*
Associate Software Engineer, Red Hat (Logging Pipeline for Distributed Storage) *Aug 2018 – June 2019*